

Roll No. :-.....

2022512

B. E. II Examination Nov-Dec 2022
Information Technology
ITR3G1: Computer Organization and Architecture

Duration: 3 Hrs

Max. Marks: 60

Total No. of Questions: 05

Note: All questions are compulsory. Attempt any two parts from each question. Make suitable assumptions if necessary. The answer should be to the point.

- Q1. A.** Give a historical perspective of computers in terms of generations and evolution of performance. 06
- B.** Define the following: 06
- | | | |
|---------------------|----------------------|----------------|
| 1. Desktop Computer | 2. Server | 3. Data Center |
| 4. Super Computer | 5. Embedded Computer | 6. Petabyte |
- C.** Write the following rules for arithmetic operations on floating-point numbers: 06
1. Add/Subtract Rule 2. Multiply Rule 3. Divide Rule
- Q2. A.** Explain caches on the processor chip. 06
- B.** How flash memory is different from EEPROM and flash drives are different from hard disk drives? 06
- C.** Consider a computer that has a cache that holds 8-word blocks and a typical DRAM main memory. 17 cycles are needed to load a block into the cache. Assume that 30 percent of the instruction in a typical program performs a read or a write operation. Assume that the hit rate in the cache is 0.95 for instructions and 0.9 for data. 1 clock cycle is needed to access information in the cache. Let us further assume that the miss penalty is the same for both read and write accesses. 06
- How effective this cache is compared to an ideal cache?
- Q3. A.** For a given CPU, the first machine cycle M1 always consists of 4 states, and the remaining machine cycles consist either of 3 or 4 states. An instruction cycle consists of a minimum of 2 and a maximum of 5 machine cycles. If the CPU's clock rate is 50 MHz and every memory access introduces one wait state. Then calculate: 06
- A). The shortest instruction execution time,
- B). The longest instruction execution time,
- B.** Design a firmware architecture for a digital component that has four states, three inputs (J, K, and Clock), and two outputs (ZERO and ONE). 06
- C.** Draw and explain the single-bus organization of the datapath inside a processor. 06